

ENGINEERING CHANGE NOTE (ECN)

PROJECT: ProjectAsynTasking

ORIGINATION: Kid Kidd

PURPOSE OF CHANGE: Round-robin cyclic executive task manager results in a more reliable design. New design will reduce labor and design costs for implementing future extensions.

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EXECUTIVE SUMMARY:

The PROJECT-ASYNC design was modified to use a round-robin cyclic executive task manager. The LED and serial operations were separated into individual tasks. A function pointer array was added to dispatch these tasks in round-robin order. The customer and high-level technical requirements remain unchanged. Testing was extended to verify the operation of the cyclic executive.

HIGH-LEVEL DESCRIPTION:

Modify PROJECT-ASYNC to use a function-pointer-based round-robin cyclic executive task manager.

REASON FOR CHANGE:

Round robin cyclic executive task manager results in a more reliable design. New design will reduce labor and design costs for implementing future extensions.

Changed Documents:

[1] kiddKid.cmpe311.fall26.project#1 - 9/26/2026

The program design, logical flow chart, source code, and testing/validation requirements were modified to implement and verify the cyclic executive task manager.

Any Additional Design Considerations:

The cyclic executive dispatches the LED 1, LED 2, and serial input tasks through a function pointer array. Each task executes briefly and returns control so the next task can execute.